

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Agile Sprint Planning & User Story Workshop
Weightage:	20%	Total Marks:	25
CO1 Statement:	<i>Apply advanced methodologies and agile project management techniques to manage software projects effectively</i>		
Student Name:	RAVURU PRANATHI	Reg. No.:	192525076
Date:	27-07-2026	Duration:	60 minutes

Code of Conduct

I R. Pranathi(192525076) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *R. Pranathi*

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

- 1. Identify the Project Vision and Stakeholders**
 - Define the project's vision, objectives, and identify all key stakeholders involved.
- 2. Create Epics and User Stories**
 - Break down the requirements into Epics and formulate well-structured User Stories using the format:
As a <user>, I want <feature>, so that <benefit>.
- 3. Define Acceptance Criteria**
 - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
- 4. Prioritize the Product Backlog**
 - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
- 5. Plan the Sprint Backlog**
 - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
- 6. Estimate Effort Using Story Points**
 - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
- 7. Present the Sprint Goal and Expected Deliverables**
 - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification	/5
Epics & User Stories	/5
Acceptance Criteria	/5
Product Backlog Prioritization & Sprint Planning	/5
Story Point Estimation, Sprint Goal & Presentation	/5
TOTAL	/25

1.Project Vision & Stakeholder Identification

Project Vision

The Smart Construction Project Management System is designed to digitally transform construction project management by replacing traditional manual processes with a centralized, cloud-based software solution. The system enables construction companies to efficiently manage multiple projects, monitor progress in real time, allocate resources effectively, track budgets, streamline procurement activities, and improve communication among all project stakeholders. By adopting Agile software development practices, the proposed solution will be flexible, scalable, secure, and capable of adapting to changing project requirements.

Business Objectives

- Reduce project completion delays by improving task scheduling.
- Minimize cost overruns through better budget monitoring.
- Automate procurement and inventory management.
- Improve communication among project teams.
- Provide real-time project monitoring through dashboards.
- Enhance decision-making using analytics and reports.
- Improve customer satisfaction by increasing project transparency.
- Support multiple construction projects simultaneously.

Project Scope

The Smart Construction Project Management System will include:

- User authentication and role management
- Project creation and planning
- Task scheduling and assignment
- Resource allocation
- Material procurement
- Budget monitoring
- Progress tracking
- Report generation
- Dashboard analytics
- Notifications and alerts
- Project closure management

Stakeholders

Stakeholder	Responsibilities	Expected Benefits
Company Management	Monitor company performance	Better decision making
Project Manager	Manage project schedule	Improved project control
Site Engineer	Update daily site activities	Accurate progress tracking
Procurement Officer	Purchase construction materials	Reduced procurement delays
Workers	Complete assigned tasks	Better work coordination
Supplier	Deliver requested materials	Faster order processing
Client	Monitor project status	Increased transparency
System Administrator	Maintain software and security	Secure and reliable system
Finance Department	Monitor project expenses	Better financial control

2. Epics & User Stories

Epic 1 – User Authentication

User Story 1

As an Admin, I want to create user accounts so that authorized users can access the system securely.

User Story 2

As an Admin, I want to assign user roles so that every user has appropriate access permissions.

Epic 2 – Project Management

User Story 3

As a Project Manager, I want to create new construction projects so that all project details are maintained in one centralized system.

User Story 4

As a Project Manager, I want to assign tasks to engineers and workers so that project activities are completed on schedule.

User Story 5

As a Project Manager, I want to monitor project milestones so that delays can be identified early.

Epic 3 – Resource Management

User Story 6

As a Site Engineer, I want to allocate workers and equipment so that available resources are utilized efficiently.

User Story 7

As a Site Engineer, I want to update daily work progress so that project status remains accurate.

Epic 4 – Material Procurement

User Story 8

As a Procurement Officer, I want to create purchase orders so that materials are delivered on time.

User Story 9

As a Supplier, I want to receive digital purchase orders so that delivery can begin immediately.

Epic 5 – Reporting

User Story 10

As Company Management, I want to view dashboards and reports so that I can monitor project performance and make better business decisions.

User Story 11

As a Client, I want to view project progress reports so that I know the current construction status.

3. Acceptance Criteria

User Story 1

Given the Admin is logged into the system

When the Admin enters valid user details

Then the new user account should be created successfully.

User Story 3

Given the Project Manager is authenticated

When project information is submitted

Then the system should save the project and generate a unique project ID.

User Story 4

Given project tasks exist

When tasks are assigned to workers

Then assigned users should receive task notifications.

User Story 6

Given available workers and equipment

When the Site Engineer allocates resources

Then resource availability should update automatically.

User Story 7

Given work has been completed

When progress is updated

Then the dashboard should immediately display the latest completion percentage.

User Story 10

Given project data exists

When Management opens the dashboard

Then project progress, budget utilization, and resource reports should be displayed.

4. Product Backlog

Priority	Product Backlog Item	Business Value	Dependency
High	User Authentication	Very High	None
High	Project Creation	Very High	Authentication
High	Task Assignment	Very High	Project Creation
High	Resource Allocation	High	Task Assignment
High	Material Procurement	High	Resource Allocation
Medium	Budget Management	Medium	Project Creation
Medium	Progress Tracking	Medium	Task Assignment
Medium	Notifications	Medium	Progress Tracking
Low	Reports	Low	Dashboard
Low	Dashboard Analytics	Low	Reports

Sprint Backlog (Sprint 1)

Task	Assigned To	Duration
Login Page	Frontend Developer	1 Day
User Authentication	Backend Developer	2 Days
Database Design	Database Engineer	2 Days
Project Creation Module	Backend Developer	3 Days
User Interface Design	UI/UX Designer	2 Days
Task Assignment Module	Full Stack Developer	3 Days
Unit Testing	QA Engineer	2 Days
Sprint Review	Scrum Team	1 Day

5. Story Point Estimation (Fibonacci)

User Story	Story Points	Complexity	Justification
US1	3	Low	Simple CRUD operations
US2	2	Low	Role assignment logic
US3	8	High	Multiple project attributes and validations
US4	5	Medium	Task scheduling logic
US5	5	Medium	Milestone tracking
US6	5	Medium	Resource allocation algorithm
US7	5	Medium	Progress update functionality
US8	8	High	Procurement workflow
US9	3	Low	Supplier notification
US10	5	Medium	Dashboard reporting
US11	3	Low	Client report viewing

Justification: User stories with more integrations, validations, and dependencies receive higher story points because they require greater development effort and testing.

Sprint Goal

The objective of Sprint 1 is to build the core foundation of the Smart Construction Project Management System by implementing user authentication, project creation, task assignment, resource allocation, and database connectivity. At the end of the sprint, users should be able to securely log in, create construction projects, assign tasks, manage resources, and monitor initial project information through the system.

Expected Sprint Deliverables

- User Registration Module
- Login and Authentication System
- Role-Based Access Control
- Project Creation Module
- Task Assignment Module
- Resource Allocation Module
- Construction Project Database
- Basic Dashboard
- Notification Module
- Sprint Review Demonstration
- Sprint Retrospective Documentation